

MCMURDO PEGASUS NORTH, AntarcticaWMO: **896670**Lat: **77.950S**Lon: **166.500E**Elev: **33**StdP: **14.68**Time Zone: **12.00 (NZL)**Period: **90-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-53.3	-49.5	N/A	N/A	N/A	N/A	N/A	N/A	53.9	-1.5	47.6	-3.1	4.3	110	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.3	36.3	N/A	33.9	N/A	31.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.0	33.6	25.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.7	23.2	10.5	-6.7	-18.6	-18.5	-19.5	-22.4	-24.3	-20.7	-7.6	12.6	24.1		
	DBStd	21.11	6.18	7.65	10.76	13.40	14.39	14.25	13.54	14.93	14.67	9.60	9.55	5.51		
	HDD50	20337	831	1105	1759	2058	2122	2084	2244	2304	2120	1784	1122	804		
	HDD65	25812	1296	1525	2224	2508	2587	2534	2709	2769	2570	2249	1572	1269		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	8.6	7.6	8.2	9.2	9.1	9.6	8.7	8.5	7.9	9.2	7.8	9.7	7.4		
	PrecAvg	6.9	0.6	0.9	0.6	0.6	0.9	1.0	0.6	0.5	0.4	0.6	0.5	0.4		
	PrecMax	13.5	2.8	2.0	2.4	1.6	3.1	5.3	2.7	1.9	1.4	2.2	1.7	0.9		
(15) Precipitation	PrecMin	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.8	0.7	0.6	0.5	0.6	0.8	1.2	0.6	0.5	0.3	0.5	0.5	0.2		
	0.4% DB	40.9	31.3	21.5	21.3	19.6	18.4	16.5	17.8	18.1	18.9	33.1	40.9			
	0.4% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	36.7	28.3	17.7	10.9	13.7	13.8	11.0	12.2	13.6	15.8	29.2	36.4			
	2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	34.4	24.8	13.6	5.6	8.5	7.6	5.4	6.1	8.4	10.8	26.8	34.1			
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% DB	32.2	21.9	10.2	1.1	2.3	1.8	-1.2	-0.6	2.2	6.6	24.7	32.2			
	10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% DB	11.9	12.4	13.0	15.8	16.6	17.5	17.0	18.3	17.1	15.7	14.2	12.3			
	5% MCDBR	15.8	15.0	13.6	17.3	18.4	19.5	20.4	20.0	17.3	18.3	14.5	15.5			
(35) Mean Daily Temperature Range	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub	0.234	0.216	0.173	0.113	N/A	N/A	N/A	N/A	0.110	0.164	0.196	0.216	0.233		
	taud	2.228	2.240	2.164	1.892	N/A	N/A	N/A	N/A	1.874	2.116	2.222	2.240	2.227		
	Ebn at Noon	306	285	242	33	0	0	0	0	28	241	294	313	314		
	Edh at Noon	31	24	15	1	0	0	0	0	1	15	25	31	34		
(44) All-Sky Solar Radiation	RadAvg	2549	1473	485	41	0	0	0	0	6	269	1126	2341	2966		
	RadStd	69	58	21	2	0	0	0	0	1	16	42	70	85		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page